BTS-VI-04.18-0674 Reg.No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

C

B.Tech. Degree VI Semester Examination April 2018

ME 15-1602 MACHINE DESIGN I

(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) What are the different steps involved in the design of a machine element?
(b) What is the difference between failure due to static load and fatigue failure?
(c) State and explain the term 'Resilience'.
(d) Explain the thread nomenclature with neat sketch.
(e) Explain the condition of self-locking of power screw.
(f) Enumerate the different types of riveted joints and rivets.
(g) Discuss the materials and practical applications for the various types of springs.
(h) What is the objective of nipping of leaf spring?
(i) What are the assumptions made in the design of welded joint?
(j) Explain the term 'critical speed of shaft'.

PART B

$(4 \times 10 = 40)$

II. A hot rolled steel shaft is subjected to a torsional moment that varies from 330 Nm clockwise to 110 Nm counter clockwise and an applied bending moment at a critical section varies from 440 Nm to 220 Nm. Determine the required shaft diameter. The material has an ultimate strength of 550 MN/m^2 and a yield strength of 410 MN/m^2. Take the endurance limit as half the ultimate strength, F.O.S = 2, size factor = 0.85, surface finish factor = 0.62, load correction factor = 1.0 in bending and 0.55 in torsion.

OR

III. A steel shaft having cylindrical cross section of yield strength 800 MPa is subjected to static loads consisting of bending moment 15 kNm and torsional moment 45 kNm. Determine the diameter of the shaft using (i) Maximum shear stress theory (ii) Maximum strain energy theory of failure. Take E = 210 GPa, Poisson's ratio = 0.25. Assume factor of safety of 2.

IV. Two 35 mm diameter shafts are connected by a flanged coupling. The flanges are fitted with 6 bolts on 125 mm bolt circle. The shafts transmit a torque of 800 Nm at 350 rpm. For the safe stresses mentioned below, calculate (i) diameter of bolts (ii) thickness of flanges (iii) key dimensions (iv) hub length (v) power transmitted.

Safe shear stress for shaft material	=	63 MPa.
Safe stress for bolt material	=	56 MPa.
Safe stress for cast iron material	=	10 MPa.
Safe stress for key material	=	46 MPa.

OR

(P.T.O.)

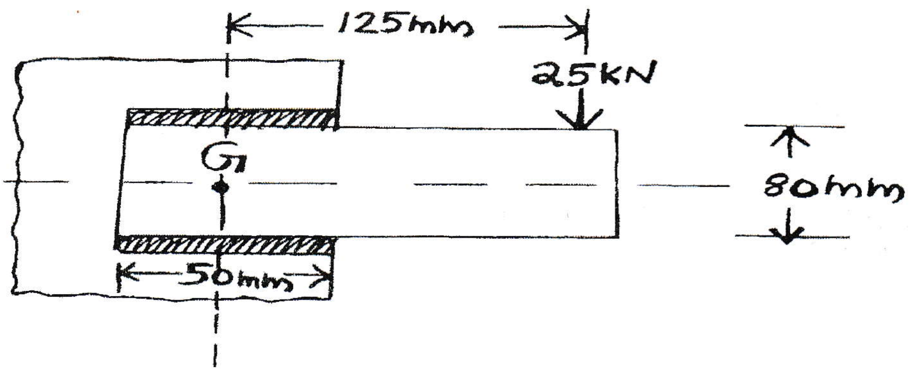
V. Design a gib and cotter joint to connect two mild steel rods for a pull of 30 kN. The maximum permissible stresses are 55 MPa in tension; 40 MPa in shear and 70 MPa in crushing. Draw a neat sketch of the joint designed.

VI. Design a triple riveted longitudinal double strap butt joint with unequal straps for a boiler. The inside diameter of the longest course of the drum is 1.3 m. The joint is to be designed for a steam pressure of 2.4 N/mm^2. The working stresses to be used are 77 MPa in tension, 62 MPa in shear and 120 MPa in crushing. Assume the efficiency of joint as 81%.

OR

VII. A helical compression spring made of oil tempered carbon steel, is subjected to a load which varies from 600 N to 1600 N. The spring index is 6 and the design factor of safety is 1.43. If the yield shear stress is 700 MPa and the endurance stress is 350 MPa, find the size of the spring wire and mean diameter of the spring coil.

VIII. A bracket carrying a load of 25 kN is to be welded as shown in figure. Find the size of weld required if the allowable shear stress is not to exceed 80 MPa.



OR

IX. A shaft made of mild steel is required to transmit 100 kW at 300 rpm. The supported length of the shaft is 3 m. It carries two pulleys each weighing 1500 N supported at a distance of 1 m from the end respectively. Assuming the safe value of stress, determine the diameter of the shaft.

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BTS–VI(R)–04.22–0289 Reg. No. | | | | | | | | |

C

B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0602 MACHINE DESIGN I

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand the customers' need, formulate the problem and draw the design specifications.
CO2: Understand component behaviour subjected to loads and identify the failure criteria.
CO3: Apply the concepts of stress analysis, theories of failure and material science to analyse, design and/or select commonly used machine components.
CO4: Analyse the stresses and strains induced in a machine element.
CO5: Design machine components using theories of failure.
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

(Use of Design Data Handbook is permitted)

(Data not given may be suitably assumed. All such assumptions must be clearly stated)

PART A

(Answer ***ALL*** questions)

			Marks	BL	CO	PO
I.		(8 × 3 = 24)				
	(a)	Discuss the statement "In static loading stress concentration in ductile materials is not as serious as in brittle material". Then why the effect of stress concentration effect has to be considered for both materials in fatigue loading?	3	L3	2	1,2
	(b)	What are the factors to be considered for selection of material used in sports equipments and for aerospace applications? Briefly explain.	3	L2	1	1,2
	(c)	List and explain the types of Keys.	3	L1	1	1
	(d)	What are the requirements of a good shaft coupling?	3	L2	2	1,2
	(e)	Explain in context of riveting: (i) Pitch (ii) Back pitch (iii) Diagonal pitch (iv) Margin.	3	L1	1	1,2
	(f)	Explain Nipping in leaf springs.	3	L2	2	1,2
	(g)	List the advantages and disadvantages of welded joints over riveted joints.	3	L2	2	1
	(h)	What are the stresses that are induced in the shafts? Briefly explain.	3	L2	2	1,2

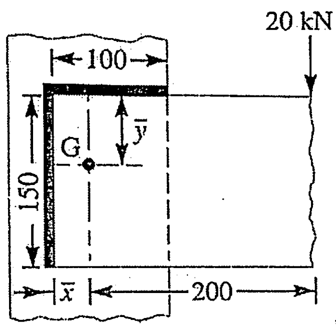
PART B

	(4 × 12 = 48)				
II.	A pulley is keyed to a shaft midway between two bearings. The shaft is made of cold drawn steel for which the ultimate strength is 550 MPa and the yield strength is 400 MPa. The bending moment at the pulley varies from -150 N-m to +400 N-m as the torque on the shaft varies from -50 N-m to +150 N-m. Obtain the diameter of the shaft for an indefinite life. The stress concentration factors for the keyway at the pulley in bending and in torsion are 1.6 and 1.3 respectively. Take the following values: Factor of safety = 1.5, Size effect factor = 0.85, Surface effect factor = 0.88	12	L5	3	2,3

OR

(P.T.O)

III.	(a)	A cylindrical shaft made of C 40 steel is subjected to static loads consisting of bending moment 10 kNm and a torsional moment 30 kNm. Determine the diameter of the shaft using any two different theories of failure, and assuming a factor of safety of 2.	6	L3	5	1,2,3
	(b)	A cantilever beam 300 mm length of circular cross section of diameter 60 mm is subjected to a load of 2 kN at its free end. The beam is also subjected to the torsional moment of 1.5 kNm. Determine the maximum principal and shear stress.	6	L3	4	2,3
IV.		Two ends of a pump are joined by means of a cotter joint at the ends. Design the joint for an axial load of 100 kN. The allowable stresses for the material used are 50 MPa in tension, 40 MPa in shear and 100 MPa in crushing.	12	L4	3	2,3
		OR				
V.		Design a flanged coupling to transmit 18 kN at 1440 rpm. The allowable shear stress for the cast iron flange is 4 MPa. The shaft, key and bolts are made of annealed steel having allowable shear stress of 93 MPa. Allowable crushing stress for the key is 186 MPa.	12	L4	3	1,2,3
VI.		Design a triple riveted double strap butt joint with chain riveting for a boiler of 1.5 m diameter and carrying a pressure of 1.2 N/mm^2. The allowable stresses are: $\sigma_t = 105$ MPa, $\tau = 77$ MPa and $\sigma_c = 162.5$ MPa	12	L3	3	2,3
		OR				
VII.		A multi leaf spring is fitted to the chassis of an automobile over a span of 1.25 m to absorb the shock due to maximum load of 24 kN. All the leaves are to be stressed to 540 MPa, when fully loaded. The spring has 2 full length leaves and 6 graduated leaves with a central band of 100 mm width. The maximum deflection is 90 mm. Determine the width and thickness of the leaf. Take E = 206 GPa.	12	L4	5	2,3
VIII.		Determine the size of the weld for plate welded as shown in figure.Take permissible shear stress for the weld material as 80 Mpa.	12	L4	3	1,2,3



All dimensions in mm.

OR

IX.	(a)	A shaft made of mild steel is required to transmit 100 kW at 300 rpm. The supported length of the shaft is 3 meters. It carries two pulleys each weighing 1500 N supported at a distance of 1 metre from the ends respectively. Assuming the safe value of stress, determine the diameter of the shaft.	8	L3	5	2,3
	(b)	Discuss the advantages and disadvantages of a hollow shaft in comparison to a solid shaft.	4	L2	2	1

Bloom's Taxonomy Levels:
L1 – 7.14% , L2 – 20.5%, L3 – 27.6%, L4 – 35.7%, L5 – 8.90%, L6 – 0%

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BTS-VI(SS)-01.19-0012 Reg. No. | | | | | | | | |

A

B.Tech. Degree VI Semester Special Supplementary Examination January 2019

ME 15-1602 MACHINE DESIGN I
(2015 Scheme)

(Use of approved design data book is permitted. Assume suitable data, if necessary)

Time: 3 Hours Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) What are the factors affecting the selection of materials?
(b) What is stress concentration? Explain the methods to reduce its effect.
(c) What is meant by creep? Discuss the design considerations.
(d) Why is an angular thread preferred in fastening?
(e) What is the special advantage of a flexible coupling over a rigid type?
(f) Briefly explain the term efficiency of a riveted joint.
(g) What is concentric helical spring and where does it find applications.
(h) What are the advantages and disadvantages of welded joints?
(i) Explain a method to obtain a bolt of uniform strength.
(j) What is meant by critical speed of shafts?

PART B

(4 × 10 = 40)

II. (a) Distinguish between static and impact loads. (2)
(b) Determine the diameter of a steel shaft using any two theories of failure subjected to a bending moment of 12 kNm and a twisting moment of 32 kNm. Assume yield strength as 800 N/mm^2, factor of safety as 3. (8)

OR

III. Determine the diameter of a circular rod made of ductile steel having ultimate tensile strength of 600 N/mm^2 and tensile yield strength of 300 N/mm^2. The member is subjected to a varying bending moment from 200 Nm clockwise and 500 Nm anticlockwise and a twisting moment varying from 50 Nm to 150 Nm. (10)
Stress concentration factor = 1.8, Factor of safety = 2

IV. A double threaded power screw is used to raise a load of 6 kN. The nominal diameter is 60 mm and the pitch is 9 mm. The threads are acme type and the coefficient of friction at the screw thread is 0.15. Neglecting collar friction, find the torque required to raise and lower the load and also efficiency of the screw for lifting load. (10)

OR

(P.T.O.)

V. Design and draw a knuckle joint made of plain carbon steel with an axial load of 90 kN. Tensile yield strength = 90 N/mm^2, shear stress = 60 N/mm^2 and crushing stress = 150 N/mm^2. (10)

VI. Design a double riveted butt joint with two cover plates for the longitudinal seam of a boiler shell 1.5 m in diameter subjected to a steam pressure of 0.9 N/mm^2. Assume joint efficiency as 75%, allowable tensile stress in the plate 90 N/mm^2, and compressive stress 150 N/mm^2 and shear stress in the rivet 60 N/mm^2. (10)

OR

VII. A locomotive semi elliptical laminated spring has an overall length of 1 m and sustains a load of 70 kN. The spring has 3 full length leaves and 15 graduated leaves with a central band of 100 mm. All the leaves are to be stressed to 400 N/mm^2 when fully loaded. The ratio of spring depth to width is 2. Determine the thickness and width of the leaves. Also find the initial gap that should be provided between the full and graduated leaves before band load is applied. (10)

VIII. (a) A 50 mm diameter solid shaft of length 200 mm is welded to a flat plate by a fillet weld along the circumference of the shaft. The shaft is subjected to a vertical load of 10 kN at its free end. Determine the size of the weld if the allowable shear stress in the weld is 100 N/mm^2. (5)

(b) A 200 × 100 × 10 mm 'angle' is welded to a frame at top and bottom. The 'angle' is subjected to a static load of 150 kN axially through the centroid of the angle plate. Determine the lengths of the weld at the top and bottom. Assume the permissible shear stress = 70 N/mm^2. (5)

OR

IX. A hollow shaft of 400 mm outside diameter and 250 mm inside diameter is supported on two bearings 3 m apart. The shaft rotates at 200 rpm and transmits 130 kW of power. The weight of the shaft is 90 kN and it receives a thrust load of 300 kN. Determine the maximum normal and shear stress induced. (10)

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BTS-VI(S.S)-11-22-1112 Reg. No. | | | | | | | | |

B

B.Tech. Degree VI Semester Special Supplementary Examination November 2022

ME 19-205-0602 MACHINE DESIGN I

(2019 *Scheme*)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand the customers' need, formulate the problem and draw the design specifications.
CO2: Understand component behaviour subjected to loads and identify the failure criteria.
CO3: Apply the concepts of stress analysis, theories of failure and material science to analyse, design and/or select commonly used machine components.
CO4: Analyse the stresses and strains induced in a machine element.
CO5: Design machine components using theories of failure.
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

(*Use of Design Data Handbook is permitted*)

(*Data not given may be suitably assumed. All such assumptions must be clearly stated*)

PART A

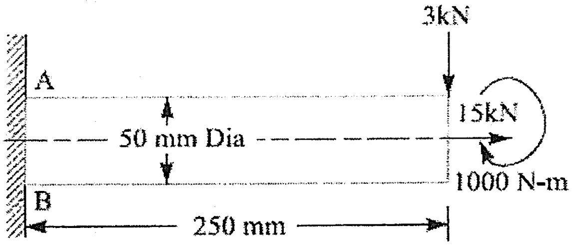
(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	What is meant by stress concentration?	3	L1	2	1
	(b)	State and explain the term 'Resilience'.	3	L2	1	1
	(c)	Give at least three practical applications of couplings.	3	L2	1	1,2
	(d)	What is meant by preloading of bolts?	3	L1	2	1
	(e)	What are various modes in which a riveted joint may fail?	3	L1	2	1,2
	(f)	Explain A.M. Wahl's factor and state its importance in designing helical springs.	3	L2	1	1,2
	(g)	Discuss the advantages of welded joints in comparison to riveted joints.	3	L2	1	1
	(h)	Explain the term: 'critical speed of shaft'.	3	L2	2	1

PART B

(4 × 12 = 48)

		Marks	BL	CO	PO
II.	A shaft, as shown in figure, is subjected to a bending load of 3 kN, pure torque of 1000 N-m and an axial pulling force of 15 kN. Calculate the stresses at A and B.	12	L4	4	1,2,3



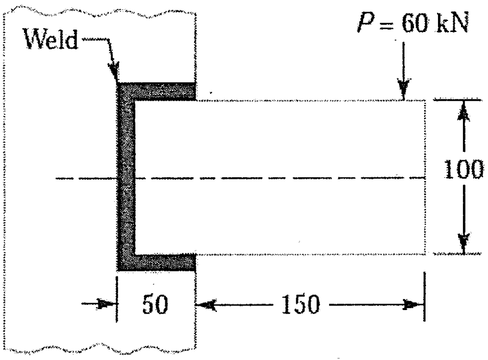
OR

(P.T.O.)

	Question	Marks	BL	CO	PO
III.	A mild steel shaft of 50 mm diameter is subjected to a bending moment of 2000 N-m and a torque T. If the yield point of the steel in tension is 200 MPa, find the maximum value of this torque without causing yielding of the shaft according to: (i) The maximum principal stress theory (ii) The maximum shear stress theory (iii) The maximum distortion strain energy theory of yielding.	12	L3	5	1,2
IV.	Design a gib and cotter joint to carry a maximum load of 35 kN. Assuming that the gib, cotter and rod are of same material and have the following allowable stresses : $\sigma_t = 20$ MPa ; $\tau = 15$ MPa ; and $\sigma_c = 50$ MPa	12	L4	3	1,2,3
	OR				
V.	Design a cast iron protective type flange coupling to transmit 15 kW at 900 rpm from an electric motor to a compressor. The service factor may be assumed as 1.35. The following permissible stresses may be used : Shear stress for shaft, bolt and key material = 40 MPa Crushing stress for bolt and key = 80 MPa Shear stress for cast iron = 8 MPa Draw a neat sketch of the coupling.	12	L4	3	1,2
VI.	Design the longitudinal joint for a 1.25 m diameter steam boiler to carry a steam pressure of 2.5 MPa. The ultimate strength of the boiler plate may be assumed as 420 MPa, crushing strength as 650 MPa and shear strength as 300 MPa. Take the joint efficiency as 80%. Sketch the joint with all the dimensions. Adopt the suitable factor of safety.	12	L3	3	1,2,3
	OR				
VII.	Design a helical spring for a spring loaded safety valve (Rams bottom safety valve) for the following conditions : Diameter of valve seat = 65 mm Operating pressure = 0.7 MPa Maximum pressure when the valve blows off freely = 0.75 MPa Maximum lift of the valve when the pressure rises from 0.7 to 0.75 MPa = 3.5 mm Maximum allowable stress = 550 MPa Modulus of rigidity = 84 GPa Spring index = 6	12	L4	5	1,2,3

(Continued)

		Marks	BL	CO	PO
VIII.	A rectangular steel plate is welded as a cantilever to a vertical column and supports a single concentrated load *P*, as shown in figure. Determine the weld size if shear stress in the same is not to exceed 140 MPa.	12	L4	3	1,2,3



OR

IX.	A shaft is supported by two bearings placed 1 m apart. A 600 mm diameter pulley is mounted at a distance of 300 mm to the right of left hand bearing and this drives a pulley directly below it with the help of belt having maximum tension of 2.25 kN. Another pulley 400 mm diameter is placed 200 mm to the left of right hand bearing and is driven with the help of electric motor and belt, which is placed horizontally to the right. The angle of contact for both the pulleys is 180° and $\mu = 0.24$. Determine the suitable diameter for a solid shaft, allowing working stress of 63 MPa in tension and 42 MPa in shear for the material of shaft. Assume that the torque on one pulley is equal to that on the other pulley.	12	L5	4	1,2,3

Bloom's Taxonomy Levels
L1 = 7.5%, L2 = 12.5%, L3 = 20.0%, L4 = 50.0%, L5 = 10.0%.

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